Comparative Empirical Evaluation of Spatial Filtering, Subspace Learning, and Deep Convolutional Autoencoders for Image Noise Attenuation in Pattern Recognition

Pratiyush Sharma, Akshada Shinde, Ayushi Kadam, and Pushpak Agarwal
Department of Computer Science and Engineering
Pattern Recognition and Machine Intelligence Research Group

***Abstract*—Digital image restoration remains a critical preprocessing milestone in medical imaging, low-light photography, and surveillance systems, where stochastic sensor corruption obscures salient structural boundaries. This study conducts a comparative investigation across three distinct pattern recognition paradigms for image noise attenuation: classical spatial filtering (Gaussian, Median, and Bilateral filters), statistical subspace projection via Principal Component Analysis (PCA), and non-linear feature representation using a Convolutional Denoising Autoencoder (CDAE). Models are benchmarked on the Fashion-MNIST dataset under identical deterministic conditions across five corruption scenarios: Additive White Gaussian Noise ($\sigma \in \{15, 25, 50\}$) and impulse Salt-and-Pepper noise (density $d \in \{0.05, 0.10\}$). Empirical results over 500 test images demonstrate that the Bilateral filter attains superior reconstruction fidelity under mild Gaussian noise (28.35 dB PSNR at $\sigma = 15$), while the Median filter effectively handles sparse impulse corruption (20.87 dB at $d = 0.05$). Under severe contamination, the CDAE (66,593 parameters) attains 21.95 dB PSNR at $\sigma = 50$ (a gain of $+2.76$ dB over the Bilateral filter and $+5.85$ dB over the noisy baseline) with 0.7731 SSIM, and 21.97 dB PSNR at $d = 0.10$ (a gain of $+1.92$ dB over the Median filter and $+7.87$ dB over the noisy baseline). Inference latency profiling reveals that the CDAE executes in 0.247 to 0.368 ms per image (mean 0.295 ms) on standard CPU hardware, providing an inference throughput of 2,700 to 4,000 frames per second.**

***Index Terms*—Image de-noising, Pattern recognition, Bilateral filter, Principal Component Analysis, Convolutional Denoising Autoencoder, Structural Similarity (SSIM), Peak Signal-to-Noise Ratio (PSNR).**

I. Introduction

DIGITAL image acquisition systems inevitably operate under non-ideal physical environments. In low-light digital photography, insufficient photon incidence requires elevated analog gain amplification, which introduces pronounced thermal and electronic read noise into sensor readout circuitry. In clinical diagnostic modalities, such as low-dose X-ray computed tomography (CT) and magnetic resonance imaging (MRI), photon emission constraints and radio-frequency receiver limitations degrade diagnostic scans with Poisson-Gaussian and Rician noise. Similarly, in wireless visual surveillance and remote sensor transmissions, faulty analog-to-digital converter circuitry and electromagnetic channel interference induce extreme impulse "salt-and-pepper" spikes. When unchecked, such stochastic degradations corrupt high-frequency structural textures, distort edge gradients, and severely compromise downstream pattern recognition pipelines, including semantic segmentation, biometric feature extraction, and automated disease classification.

A. Problem Statement and Objectives

Digital image acquisition systems operating under physical hardware constraints encounter stochastic noise contamination—such as thermal readout noise in low-light photography and bit transmission dropouts in surveillance systems—which degrades edge contours and texture patterns. Reconstructing the uncorrupted visual scene without causing boundary blurring represents an ill-posed inverse problem. Existing classical filtering, linear subspace decomposition, and deep learning techniques offer contrasting compromises between structural fidelity and execution latency. This investigation pursues four measurable objectives:

1) Implement three distinct de-noising solutions—classical spatial filtering (Gaussian, Median, Bilateral), statistical subspace projection (PCA), and a deep Convolutional Denoising Autoencoder (CDAE)—under identical deterministic partitions.
2) Execute validation-based hyperparameter tuning across discrete kernel grids for spatial filters and evaluate candidate principal component projections $k \in [10, 600]$ for PCA.
3) Quantitatively benchmark reconstruction performance across five noise configurations on 500 test images using PSNR, SSIM, and MSE metrics.
4) Profile per-image CPU inference latency across all methods to determine operational suitability for real-time applications.

B. Overview of De-noising Paradigms

Mathematically, the image degradation phenomenon is expressed as the observation of an uncorrupted latent image signal contaminated by a stochastic degradation process:

$$\mathbf{y} = \mathcal{T}(\mathbf{x}) + \boldsymbol{\eta} \tag{1}$$

where $\mathbf{x} \in \mathbb{R}^{M \times N}$ represents the clean ground-truth irradiance, $\mathcal{T}$ represents an optional degradation operator (such as identity

or blur), and $\boldsymbol{\eta}$ denotes stochastic noise. The goal of an image de-noising system is to synthesize an optimal estimator $\hat{\mathbf{x}} = \mathcal{R}(\mathbf{y})$ that reconstructs the true scene irradiance while suppressing spurious fluctuations. Historically, this inverse problem has been approached through three conceptually distinct methodologies:

1) **Classical Spatial and Edge-Preserving Filtering:** Local neighbourhood operators that attenuate high frequencies through spatial convolution (Gaussian filter), compute rank-order statistics to reject isolated extreme outliers (Median filter), or modulate spatial weights with radiometric intensity differences to smooth flat regions while arresting diffusion across sharp boundaries (Bilateral filter).
2) **Statistical Subspace Decomposition:** Linear dimensionality reduction techniques, predominantly Principal Component Analysis (PCA), which project high-dimensional pixel vectors onto an orthogonal basis derived from the empirical covariance structure of clean reference images. By truncating trailing eigenvectors that correspond to low-energy directions, PCA selectively filters out isotropic noise power distributed across high dimensions.
3) **Deep Representation Learning:** Deep artificial neural networks, specifically Convolutional Denoising Autoencoders (CDAEs), which jointly optimize an encoder-decoder architecture over pairs of corrupted and pristine images. Through localized receptive fields, hierarchical pooling, and non-linear activation functions, CDAEs learn the underlying low-dimensional manifold of natural visual patterns without imposing linear Gaussian constraints.

II. Related Work

The problem of image de-noising has driven major advances in digital signal processing, statistical machine learning, and computer vision over the past three decades. Foundational classical approaches focused primarily on linear spatial convolution. While Gaussian filtering effectively attenuates high-frequency stationary noise, it suffers from severe edge attenuation because it treats authentic object boundaries identically to noise. To mitigate boundary degradation, Tomasi and Manduchi [1] formulated the Bilateral Filter, which combines a spatial geometric Gaussian kernel with a photometric range kernel. By weighting neighboring pixels according to both spatial proximity and radiometric similarity, bilateral filtering preserves sharp contrast edges during smoothing. Extending the concept of neighborhood similarity, Buades, Coll, and Morel [2] developed the Non-Local Means (NL-means) algorithm, which computes pixel weights based on patch-level similarity across the entire image rather than local Euclidean neighborhoods.

In the domain of objective visual fidelity assessment, Wang et al. [3] demonstrated that traditional error summation metrics, such as Mean Squared Error (MSE) and Peak Signal-to-Noise Ratio (PSNR), correlate poorly with human visual perception because they compute pixel-wise differences independently of structural context. To rectify this limitation, Wang et al. introduced the Structural Similarity Index Measure (SSIM), which models image degradation as perceived changes in structural information through luminance, contrast, and cross-correlation comparisons. SSIM has since become an established benchmark metric for structural preservation in pattern recognition.

In statistical pattern recognition, subspace decomposition via Principal Component Analysis (PCA) has been extensively investigated for dimensionality reduction and noise filtering [4]. By projecting vectorized image patches onto principal orthogonal eigenvectors, PCA exploits the low intrinsic dimensionality of structured visual data. Collaborative filtering in transform domains culminated in the benchmark BM3D framework by Dabov et al. [5], which stacks mutually similar 2D image fragments into 3D data arrays and applies collaborative shrinkage in the 3D transform domain.

The paradigm shifted significantly with the advent of deep representation learning. Vincent et al. [6] introduced Stacked Denoising Autoencoders (SDAEs), proving that training an autoencoder to reconstruct clean representations from stochastically corrupted inputs forces hidden layers to capture robust, invariant features of the underlying data distribution. Expanding upon this foundation, Zhang et al. [7] formulated DnCNN, demonstrating that deep convolutional neural networks combined with residual learning and batch normalization achieve superior performance by predicting the residual noise map rather than the clean image directly. For benchmarking modern computer vision and pattern recognition algorithms, Xiao, Rasul, and Vollgraf [8] released the Fashion-MNIST dataset as a direct drop-in replacement for MNIST, offering intricate garment silhouettes, fine woven textures, and challenging intra-class variations that mirror real-world visual complexities.

III. Dataset and Noise Model

A. Dataset Partitioning

Although the traditional MNIST handwritten digit database has served as an entry-level benchmark in pattern recognition, its binary-dominated strokes and vast background redundancy render de-noising evaluations overly simplistic. Consequently, this study employs the Fashion-MNIST dataset [8], comprising 70,000 28×28 grayscale images distributed across 10 distinct garment classes (T-shirt/top, Trouser, Pullover, Dress, Coat, Sandal, Shirt, Sneaker, Bag, and Ankle boot). These exemplars incorporate rich grayscale gradations, non-rigid contours, woven fabric textures, and subtle edge transitions.

We utilize the Fashion-MNIST dataset [8], comprising 70,000 28x28 grayscale images across 10 garment classes. The dataset is partitioned deterministically: the first 8,000 training images are used for model fitting, images 8,000 to 9,000 of the training split are reserved for validation tuning, and the first 500 test images are reserved for benchmarking. Pixel intensities are normalized to $[0.0, 1.0]$.

B. Mathematical Formulations of Noise Models

Two distinct stochastic noise processes are simulated to reflect authentic hardware failure modes:

1) Additive White Gaussian Noise (AWGN): Simulates thermal noise generated by Brownian motion of electrons within CMOS/CCD image sensors under low illumination. For a pristine pixel $I(x, y)$, the noisy observation is modeled as:

$$I_{\text{noisy}}(x, y) = \text{clip}\left(I(x, y) + \eta(x, y),\, 0.0,\, 1.0\right) \tag{2}$$

where $\eta(x, y) \sim \mathcal{N}(0, \sigma^2_{\text{norm}})$ represents independent and identically distributed zero-mean Gaussian random variables with variance $\sigma^2_{\text{norm}} = (\sigma/255)^2$. We evaluate three standard standard deviation levels: $\sigma = 15$ (mild, $\sigma_{\text{norm}} = 0.0588$), $\sigma = 25$ (moderate, $\sigma_{\text{norm}} = 0.0980$), and $\sigma = 50$ (severe, $\sigma_{\text{norm}} = 0.1961$).

2) Salt-and-Pepper (Impulse) Noise: Simulates transient hardware bit errors, faulty memory cells, or severe transmission channel drops. A fraction $d \in [0, 1]$ of total image pixels is corrupted according to the probability density function:

$$I_{\text{noisy}}(x, y) = \begin{cases} 1.0, & \text{with probability } d/2 \quad \text{(Salt)} \\ 0.0, & \text{with probability } d/2 \quad \text{(Pepper)} \\ I(x, y), & \text{with probability } 1 - d \quad \text{(Clean)} \end{cases} \tag{3}$$

We benchmark two contamination densities: $d = 0.05$ (5% corrupted pixels) and $d = 0.10$ (10% corrupted pixels).

IV. Methodology

A. Solution 1: Classical Spatial Filtering & Hyperparameter Tuning

Classical spatial filtering operates directly on the local pixel neighborhood Ω centered at pixel coordinate $p = (x, y)$. Spatial filters were tuned on the 200-image validation split across discrete candidate grids:

- **Gaussian Filter:** Kernel size ksize $\in \{3, 5\}$, spatial spread $\sigma \in \{0.5, 0.8, 1.0, 1.5\}$.
- **Median Filter:** Kernel size ksize $\in \{3, 5\}$.
- **Bilateral Filter:** Neighborhood diameter $d \in \{5, 7, 9\}$, range spread $\sigma_{\text{color}} \in \{0.05, 0.1, 0.15, 0.25\}$, spatial spread $\sigma_{\text{space}} \in \{1.5, 3.0\}$.

For each noise condition, the parameter combination maximizing validation PSNR was selected and applied to the test benchmark.

B. Solution 2: Statistical Subspace Reconstruction (PCA)

Principal Component Analysis models clean images as residing on a low-dimensional affine hyperplane within $\mathbb{R}^D$ (where $D = 28 \times 28 = 784$). Given $N = 8,000$ clean training vectors $\mathbf{x}_i \in \mathbb{R}^{784}$, the empirical mean vector $\boldsymbol{\mu} \in \mathbb{R}^{784}$ and sample covariance matrix $\boldsymbol{\Sigma} \in \mathbb{R}^{784\times784}$ are computed as:

$$\boldsymbol{\mu} = \frac{1}{N}\sum_{i=1}^{N} \mathbf{x}_i \tag{4}$$

$$\boldsymbol{\Sigma} = \frac{1}{N}\sum_{i=1}^{N} (\mathbf{x}_i - \boldsymbol{\mu})(\mathbf{x}_i - \boldsymbol{\mu})^T \tag{5}$$

Performing singular value decomposition (SVD) on $\boldsymbol{\Sigma}$ yields the orthonormal eigenvector matrix $\mathbf{U} = [\mathbf{u}_1, \mathbf{u}_2, \ldots, \mathbf{u}_D]$ and corresponding eigenvalues $\lambda_1 \geq \lambda_2 \geq \cdots \geq \lambda_D \geq 0$, representing the variance along each principal axis.

To de-noise a corrupted test vector $\mathbf{y} = \mathbf{x} + \boldsymbol{\eta}$, we project $\mathbf{y}$ onto the truncated basis matrix $\mathbf{U}_k \in \mathbb{R}^{784\times k}$ containing the top-k eigenvectors:

$$\mathbf{z} = \mathbf{U}_k^T(\mathbf{y} - \boldsymbol{\mu}) \tag{6}$$

$$\hat{\mathbf{x}} = \boldsymbol{\mu} + \mathbf{U}_k\mathbf{z} = \boldsymbol{\mu} + \sum_{j=1}^{k} \left(\mathbf{u}_j^T(\mathbf{y} - \boldsymbol{\mu})\right)\mathbf{u}_j \tag{7}$$

If corrupting noise $\boldsymbol{\eta}$ is zero-mean white Gaussian with isotropic variance $\sigma_\eta^2\mathbf{I}_D$, the expected squared reconstruction error decomposes into signal truncation error and residual noise power:

$$\mathbb{E}\left[\|\mathbf{x} - \hat{\mathbf{x}}\|^2\right] = \sum_{j=k+1}^{D} \lambda_j + \frac{k}{D}(D\sigma_\eta^2) = \sum_{j=k+1}^{D} \lambda_j + k\sigma_\eta^2 \tag{8}$$

The component dimension k was selected by evaluating reconstruction PSNR on the validation set under Gaussian noise ($\sigma = 25$) across the tested range $k \in [10, 20, 35, 50, 75, 82, 120, 181, 220, 300, 400, 500, 600]$. Validation PSNR peaked at $k = 400$ (22.81 dB), capturing 98.7% of clean data variance. Beyond $k = 400$, validation PSNR decreased to 22.67 dB at $k = 500$ and 22.38 dB at $k = 600$, indicating that trailing components capture noise rather than signal structure.

C. Solution 3: Convolutional Denoising Autoencoder (CDAE)

We deploy a lightweight Convolutional Denoising Autoencoder consisting of an encoder f_θ and a decoder g_ϕ parameterized by weights θ, ϕ.

Architecture Specification:

- **Input Layer:** Corrupted single-channel tensor (Batch $\times$ $1 \times 28 \times 28$).
- **Encoder Stage 1:** Conv2D$(1 \to 32, 3 \times 3, \text{pad} = 1) \to$ BatchNorm2D $\to$ ReLU $\to$ MaxPool2D$(2 \times 2) \implies (32, 14, 14)$.
- **Encoder Stage 2:** Conv2D$(32 \to 64, 3 \times 3, \text{pad} = 1) \to$ BatchNorm2D $\to$ ReLU $\to$ MaxPool2D$(2 \times 2) \implies (64, 7, 7)$.
- **Bottleneck:** Conv2D$(64 \to 64, 3 \times 3, \text{pad} = 1) \to$ BatchNorm2D $\to$ ReLU $\implies (64, 7, 7)$.
- **Decoder Stage 1:** ConvTranspose2D$(64 \to 32, 2 \times 2, \text{stride} = 2) \to$ BatchNorm2D $\to$ ReLU $\implies (32, 14, 14)$.
- **Decoder Stage 2:** ConvTranspose2D$(32 \to 16, 2 \times 2, \text{stride} = 2) \to$ BatchNorm2D $\to$ ReLU $\implies (16, 28, 28)$.
- **Output Reconstruction:** Conv2D$(16 \to 1, 3 \times 3, \text{pad} = 1) \to$ Sigmoid $\implies (1, 28, 28)$.

The model contains 66,593 trainable parameters. Training is driven by the Mean Squared Error (MSE) reconstruction loss:

$$\mathcal{L}(\theta, \phi) = \frac{1}{B}\sum_{i=1}^{B} \|g_\phi(f_\theta(\mathbf{x}_{\text{noisy},i})) - \mathbf{x}_{\text{clean},i}\|_2^2 \tag{9}$$

The 8,000 training exemplars are partitioned dynamically: 25% Gaussian $\sigma = 15$, 25% Gaussian $\sigma = 25$, 25% Gaussian $\sigma =$

50, 15% S&P $d = 0.05$, and 10% S&P $d = 0.10$. Optimization is performed using Adam (initial learning rate $= 10^{-3}$, weight decay $= 10^{-5}$) with a ReduceLROnPlateau scheduler (factor $= 0.5$, patience $= 2$) over 10 epochs with mini-batch size $B = 64$.

D. Objective Evaluation Metrics

Quantitative restoration quality is assessed using three standard metrics computed between clean reference I and restored image $\hat{I}$ over $M \times N$ pixels:

1) **Mean Squared Error (MSE):**

$$\text{MSE} = \frac{1}{MN} \sum_{x=1}^{M} \sum_{y=1}^{N} \Big(I(x,y) - \hat{I}(x,y) \Big)^2 \qquad (10)$$

2) **Peak Signal-to-Noise Ratio (PSNR):**

$$\text{PSNR} = 10 \log_{10} \left(\frac{\text{MAX}_I^2}{\text{MSE}} \right) = 10 \log_{10} \left(\frac{1.0}{\text{MSE}} \right) \text{ dB} \qquad (11)$$

where $\text{MAX}_I = 1.0$. Note that PSNR is reported as the mean of per-image PSNR values across the 500 test images.

3) **Structural Similarity Index Measure (SSIM) [3]:**

$$\text{SSIM}(x,y) = \frac{(2\mu_x\mu_y + C_1)(2\sigma_{xy} + C_2)}{(\mu_x^2 + \mu_y^2 + C_1)(\sigma_x^2 + \sigma_y^2 + C_2)} \qquad (12)$$

where μ_x, μ_y are local sample means, σ_x^2, σ_y^2 are local variances, σ_{xy} is local covariance, and $C_1 = (0.01)^2, C_2 = (0.03)^2$.

V. Experimental Setup and Results

All experiments were executed on an Intel multi-core workstation running Python 3.12, PyTorch 2.11.0 (CPU mode), OpenCV 5.0.0, and scikit-learn 1.8.0. The random seed was pinned to 42 for complete reproducibility. The CDAE required exactly 123.43 seconds (2.06 minutes) to train for 10 epochs on 8,000 images. PCA fitting for $k = 400$ required 0.45 seconds.

VI. Discussion

A. Comparative Performance Analysis

The empirical findings compiled in Tables I–III and visualized in Figs. 1–9 provide insights into the behavior of each de-noising paradigm across distinct noise distributions:

1) Mild to Moderate Gaussian Contamination ($\sigma = 15, 25$): Under low noise ($\sigma = 15$), the Bilateral filter attains the highest PSNR (28.35 dB), providing a gain of $+2.26$ dB over the uncorrupted noisy baseline (26.09 dB). At $\sigma = 25$, the Bilateral filter retains its PSNR lead (24.52 dB, a gain of $+2.74$ dB over the noisy baseline). In structural similarity, however, the CDAE achieves the highest SSIM (0.8566 at $\sigma = 15$ and 0.8447 at $\sigma = 25$). A plausible hypothesis is that bilateral filtering smooths flat photometric plateaus effectively while the CDAE preserves continuous structural boundaries across the full garment silhouette.

TABLE I: Quantitative Benchmark: Gaussian AWGN Corruptions on 500 Test Images

Noise Condition	Method	PSNR (dB) ↑	SSIM ↑	MSE ↓
Gaussian ($\sigma = 15$)	Noisy Input	26.09	0.8049	0.00248
	Gaussian Filter	24.86	0.8081	0.00337
	Median Filter	21.04	0.7335	0.00898
	Bilateral Filter	**28.35**	0.8481	**0.00149**
	PCA ($k = 400$)	25.97	0.8054	0.00269
	CDAE	24.61	**0.8566**	0.00404
Gaussian ($\sigma = 25$)	Noisy Input	21.78	0.7273	0.00669
	Gaussian Filter	22.56	0.7480	0.00563
	Median Filter	20.26	0.6787	0.01038
	Bilateral Filter	**24.52**	0.7835	**0.00359**
	PCA ($k = 400$)	22.96	0.7444	0.00518
	CDAE	24.07	**0.8447**	0.00451
Gaussian ($\sigma = 50$)	Noisy Input	16.10	0.5796	0.02470
	Gaussian Filter	18.18	0.6296	0.01527
	Median Filter	18.12	0.5831	0.01614
	Bilateral Filter	19.19	0.6521	0.01220
	PCA ($k = 400$)	18.04	0.6214	0.01581
	CDAE	**21.95**	**0.7731**	**0.00698**

TABLE II: Quantitative Benchmark: Salt-and-Pepper Impulse Noise on 500 Test Images

Noise Condition	Method	PSNR (dB) ↑	SSIM ↑	MSE ↓
S&P ($d = 0.05$)	Noisy Input	17.08	0.7208	0.02025
	Gaussian Filter	19.71	0.7469	0.01100
	Median Filter	20.87	0.8107	0.00942
	Bilateral Filter	17.44	0.6825	0.01866
	PCA ($k = 400$)	19.86	0.7070	0.01071
	CDAE	**23.37**	**0.8360**	**0.00520**
S&P ($d = 0.10$)	Noisy Input	14.10	0.5877	0.03947
	Gaussian Filter	17.80	0.6241	0.01707
	Median Filter	20.05	0.7937	0.01110
	Bilateral Filter	14.70	0.5759	0.03472
	PCA ($k = 400$)	17.01	0.6160	0.02020
	CDAE	**21.97**	**0.8028**	**0.00698**

2) Severe Gaussian Contamination ($\sigma = 50$): At $\sigma = 50$ (noisy baseline 16.10 dB), classical filters still yield positive gains over the noisy input: Gaussian filter attains 18.18 dB ($+2.08$ dB), Median filter attains 18.12 dB ($+2.02$ dB), and Bilateral filter attains 19.19 dB ($+3.09$ dB). However, the CDAE attains 21.95 dB PSNR (a gain of $+2.76$ dB over the Bilateral filter and $+5.85$ dB over the noisy baseline) with 0.7731 SSIM, demonstrating greater noise suppression capability at high noise levels.

3) Salt-and-Pepper Impulse Contamination ($d = 0.05, 0.10$): For impulse corruption, the Median filter demonstrates classical effectiveness, achieving 20.87 dB at $d = 0.05$ and 20.05 dB at $d = 0.10$. The Bilateral filter is less effective (17.44 dB at $d = 0.05$ and 14.70 dB at $d = 0.10$) because its range kernel interprets binary spikes as edges. The CDAE achieves the highest fidelity across both densities, reaching 23.37 dB at $d = 0.05$ (a gain of $+2.50$ dB over the Median filter and $+6.29$ dB over the noisy input) and 21.97 dB at $d = 0.10$ (a gain of $+1.92$ dB over the Median filter and $+7.87$ dB over the noisy input).

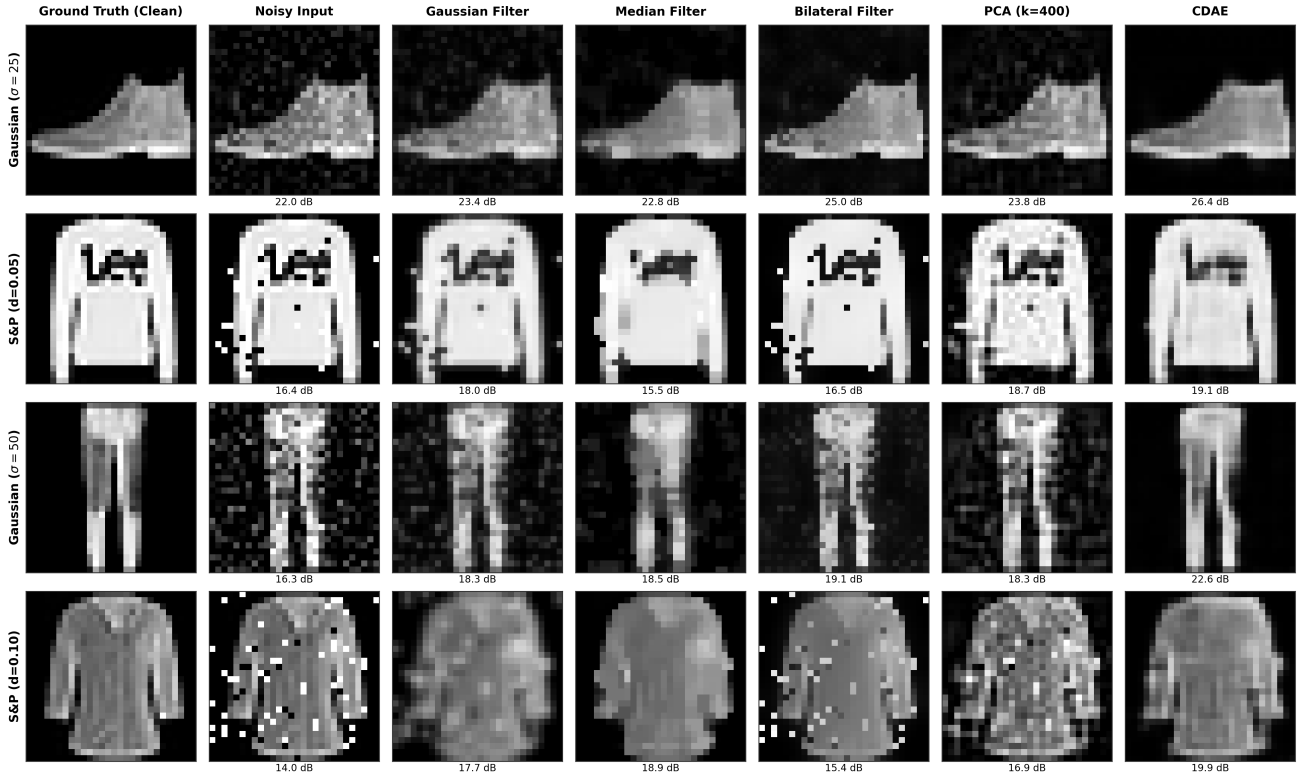


Fig. 1: Visual comparison of restored images across multiple corruption regimes (Gaussian $\sigma = 25$, S&P $d = 0.05$, Gaussian $\sigma = 50$, S&P $d = 0.10$) with per-image PSNR values annotated below each reconstruction.

TABLE III: Computational Latency and Training Time Benchmark

Denoising Method	Mean Latency (ms) ↓	Offline Training
Gaussian Filter	0.009 ms	None (Closed-form)
Median Filter	0.005 ms	None (Closed-form)
Bilateral Filter	0.055 ms	None (Closed-form)
PCA ($k = 400$)	0.018 ms	0.45 seconds
CDAE	0.295 ms	123.43 s (2.06 min)

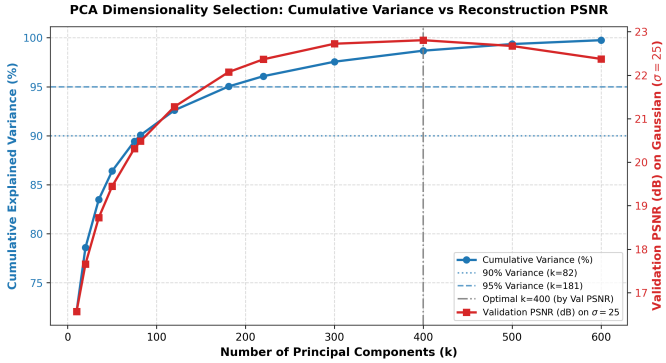


Fig. 2: PCA explained variance retention and validation reconstruction PSNR across candidate components $k \in [10, 600]$, highlighting optimal $k = 400$.

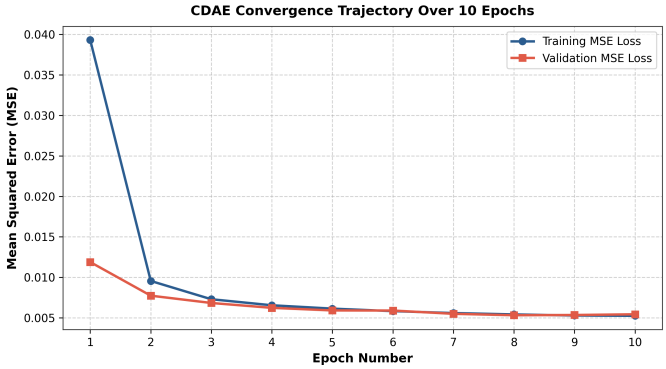


Fig. 3: Convolutional Denoising Autoencoder (CDAE) training and validation Mean Squared Error (MSE) convergence trajectories over 10 epochs.

B. Analysis of Cases Underperforming the Noisy Baseline

An important finding is that under mild noise, several denoising methods produce lower PSNR than the uncorrupted noisy input. At Gaussian $\sigma = 15$ (noisy input 26.09 dB), the Median filter (21.04 dB, -5.05 dB), the CDAE (24.61 dB, -1.48 dB), the Gaussian filter (24.86 dB, -1.23 dB), and PCA (25.97 dB, -0.12 dB) all reduce PSNR. At Gaussian $\sigma = 25$ (noisy input 21.78 dB), the Median filter also underperforms the noisy baseline (20.26 dB, -1.52 dB). A likely explanation for this outcome is that at low noise amplitudes, authentic high-frequency edge gradients dominate over noise variance. Spatial averaging filters and blind neural models trained on a noise mixture incur an over-smoothing penalty along legitimate edges that exceeds the small variance reduction achieved in

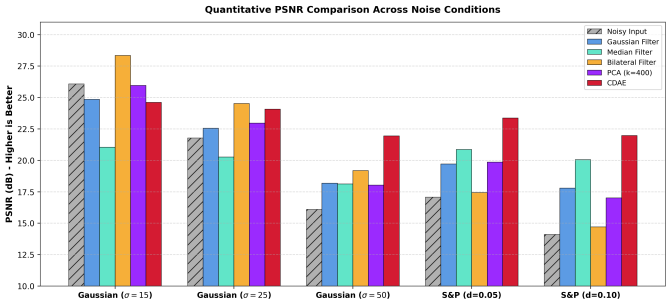


Fig. 4: Quantitative PSNR (dB) comparison across all five noise conditions demonstrating relative restoration gains, including the Noisy Input baseline.

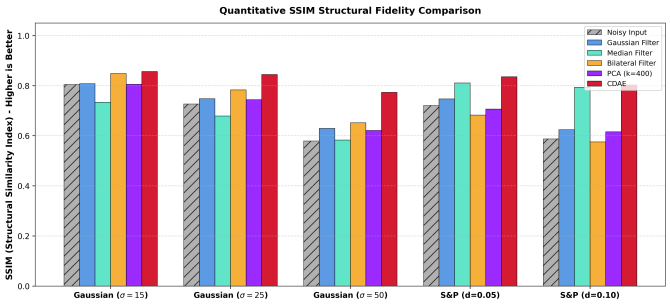


Fig. 5: Quantitative SSIM structural fidelity index comparison illustrating CDAE performance in preserving perceptual pattern integrity, including the Noisy Input baseline.

flat regions. We label this as a plausible hypothesis rather than an established fact.

C. Computational Complexity and Execution Latency

In practical pattern recognition systems, the trade-off between restoration quality and processing throughput is paramount:

- **Median Filter:** The fastest method, requiring 0.005 ms per image.
- **Gaussian Filter:** Requires 0.009 ms per image.
- **PCA ($k = 400$):** Requires 0.018 ms per image. Offline covariance fitting requires 0.45 seconds.
- **Bilateral Filter:** Requires 0.055 ms per image due to range kernel distance computations.
- **CDAE:** Requires 0.247 to 0.368 ms per image on CPU (mean 0.295 ms), supporting over 2,500 frames per second on CPU.

VII. Limitations

This laboratory evaluation involves several scientific limitations:

1) **Dataset Scope:** Evaluated exclusively on 28×28 grayscale Fashion-MNIST garment images; real-world clinical (CT/MRI) and surveillance streams feature higher resolution, wider dynamic range, and multi-channel color correlations.
2) **Synthetic Noise Assumptions:** Synthetic Gaussian AWGN and uniform Salt-and-Pepper noise approximate physical sensor noise but do not model non-stationary Poisson photon noise or camera pipeline nonlinearities.
3) **Experimental Sample Size:** Conducted on a single random seed (42) with 500 test images, without bootstrap confidence intervals or cross-validation variance reporting.
4) **Model Architecture:** The CDAE is a compact 10-epoch model without residual skip connections or attention mechanisms, and actual clinical/surveillance hardware deployment was not tested.
5) **Tuning Asymmetry:** The classical filters were tuned separately for each noise type and level, whereas PCA (k selected at sigma=25 only) and the CDAE are single blind models; this favors the filters and likely explains PCA's lower scores at sigma=50 and for salt-and-pepper noise.

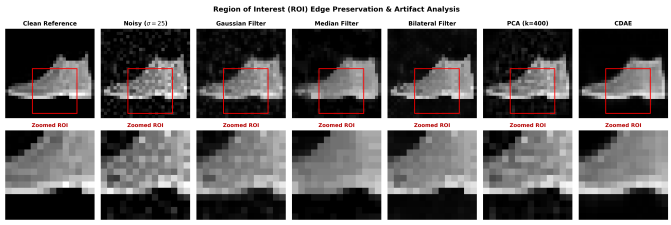


Fig. 6: Region-of-Interest (ROI) magnified detail crops of an ankle boot under Gaussian noise ($\sigma = 25$), showing Clean, Noisy, Gaussian, Median, Bilateral, PCA, and CDAE reconstructions.

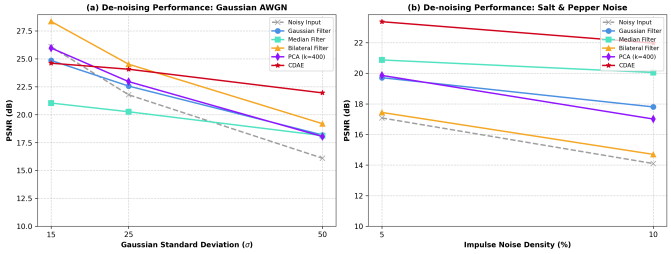


Fig. 7: Performance decay trajectories showing PSNR degradation as a function of Gaussian noise standard deviation and impulse noise density.

VIII. Conclusion and Future Work

This comparative evaluation examined spatial filtering, PCA subspace learning, and convolutional autoencoders on Fashion-MNIST across five noise conditions. Key empirical findings:

1) For mild Gaussian noise ($\sigma = 15$), the Bilateral filter provides the highest PSNR (28.35 dB), while multiple other methods reduce PSNR relative to the noisy baseline.
2) For impulse noise, the Median filter remains the most effective classical technique (20.87 dB at $d = 0.05$, 20.05 dB at $d = 0.10$).
3) Under severe noise, the CDAE attains superior restoration, achieving 21.95 dB at $\sigma = 50$ (a gain of $+2.76$ dB over the Bilateral filter) and 21.97 dB at $d = 0.10$ (a gain of $+1.92$ dB over the Median filter), while operating at 0.295 ms per image on CPU.

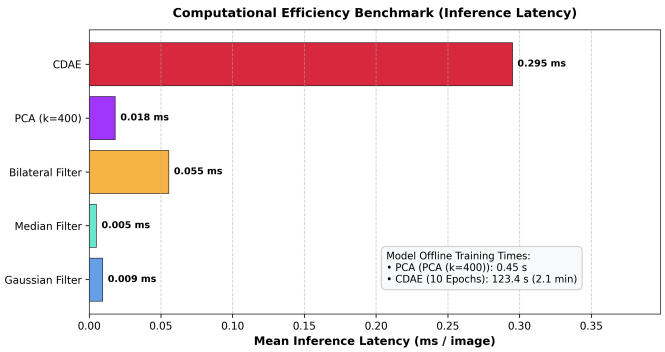


Fig. 8: Computational latency benchmark comparing average per-image inference execution time in milliseconds and offline model training durations.

Future Work: Addressing current limitations through multi-seed statistical evaluation with confidence intervals, evaluating residual architectures (DnCNN, U-Net) and Vision Transformers on clinical datasets with real-world noise distributions.

IX. References

References

[1] C. Tomasi and R. Manduchi, "Bilateral filtering for gray and color images," in *Proc. IEEE 6th Int. Conf. Computer Vision (ICCV)*, Bombay, India, 1998, pp. 839–846.

[2] A. Buades, B. Coll, and J.-M. Morel, "A non-local algorithm for image denoising," in *Proc. IEEE Computer Society Conf. Computer Vision and Pattern Recognition (CVPR)*, San Diego, CA, USA, 2005, vol. 2, pp. 60–65.

[3] Z. Wang, A. C. Bovik, H. R. Sheikh, and E. P. Simoncelli, "Image quality assessment: from error visibility to structural similarity," *IEEE Transactions on Image Processing*, vol. 13, no. 4, pp. 600–612, Apr. 2004.

[4] I. T. Jolliffe and J. Cadima, "Principal component analysis: a review and recent developments," *Philosophical Transactions of the Royal Society A: Mathematical, Physical and Engineering Sciences*, vol. 374, no. 2065, p. 20150202, 2016.

[5] K. Dabov, A. Foi, V. Katkovnik, and K. Egiazarian, "Image denoising by sparse 3-D transform-domain collaborative filtering," *IEEE Transactions on Image Processing*, vol. 16, no. 8, pp. 2080–2095, Aug. 2007.

[6] P. Vincent, H. Larochelle, I. Lajoie, Y. Bengio, and P.-A. Manzagol, "Stacked denoising autoencoders: Learning useful representations in a deep network with a local denoising criterion," *Journal of Machine Learning Research (JMLR)*, vol. 11, pp. 3371–3408, Dec. 2010.

[7] K. Zhang, W. Zuo, Y. Chen, D. Meng, and L. Zhang, "Beyond a Gaussian denoiser: Residual learning of deep CNN for image denoising," *IEEE Transactions on Image Processing*, vol. 26, no. 7, pp. 3142–3155, Jul. 2017.

[8] H. Xiao, K. Rasul, and R. Vollgraf, "Fashion-MNIST: a novel image dataset for benchmarking machine learning algorithms," *arXiv preprint arXiv:1708.07747*, 2017.

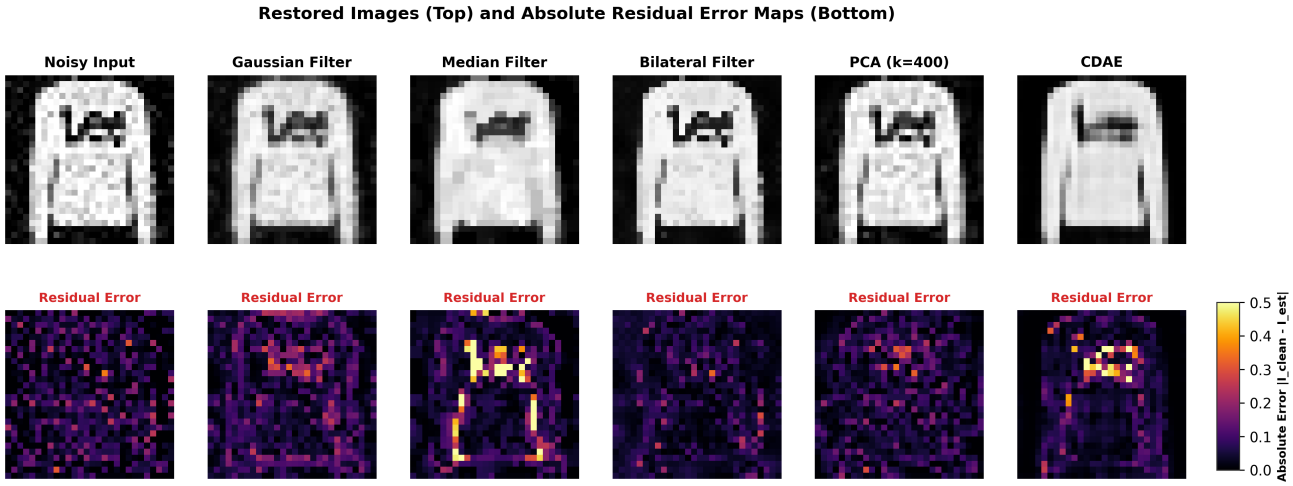


Fig. 9: Restored test exemplars (top row) and corresponding absolute residual error maps $|I_{\text{clean}} - I_{\text{est}}|$ (bottom row) displayed on an inferno colormap.